Biochemistry Cume
November 21, 2008

Total points: 100

1. In an experiment, soluble extracts of *Escherichia coli* were incubated with the 2'- deoxynucleotides dATP, dTTP, dGTP and dCTP, all labeled with ^{32}P in the α-phosphate group. After a fixed time, the incubation mixture was treated with trichloroacetic acid (TCA), which precipitates the DNA but not the deoxynucleotide precursors. The extent of incorporation of the deoxynucleotide precursors was determined from the amount of radioactivity present in the TCA-induced precipitate. With this background,

   a. Would you expect radioactivity to be found in the precipitate if any one of the four nucleotide precursors were omitted from the incubation mixture? Explain. (**15 points**)

   b. Would ^{32}P label be incorporated into the DNA if only dTTP were labeled? Explain. (**15 points**)

   c. Would you find radioactivity in the TCA-induced precipitate, if precursor deoxynucleotides containing ^{32}P labels at the β or γ phosphate were to be used in the reaction? Explain. (**15 points**)

2. The *E.coli* chromosome contains ~ 4.64×10^6 base pairs.

   a. How many turns of the double helix must be unwound during the replication process? (**20 points**)

   b. Chromosomal replication of *E.coli* starts at a fixed origin and proceeds bidirectionally. How long would it take to replicate the *E.coli* chromosome at 37 °C if two replication forks proceed from the origin? Assume replication occurs at the rate of 1000 base pairs per second. (**15 points**)

3. An *E.coli* mutant, defective in DNA ligase, was grown in a medium containing ^3H-labeled thymine. Alkaline sucrose gradient of the actively replicating mutant *E.coli* DNA produced two distinct radioactive bands. What do these bands represent? Explain. (**20 points**)